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WATER-GAS SHIFT REACTION OVER AN IRON OXIDE/CHROMIUM OXIDE CATALYST. I: MASS TRANSPORT EFFECTS

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ABSTRACT

Investigation of the water-gas shift reaction in a microreactor and in a semi technical plant at pressures up to 30 bar over a commercial iron oxide/chromia catalyst has allowed the following conclusions to be drawn.

The true activation energy for the reaction is about 121 kJ mol^{-1}.

The reaction becomes significantly pore diffusion limited over 5.4 mm pellets at temperatures above 370°C and at temperatures above 350°C over 8.5 mm pellets at 30 bar operating pressure.

A detailed assessment of heat and mass transfer effects in the microreactor is given.

INTRODUCTION

The water-gas shift reaction, the reaction of steam with carbon monoxide to produce carbon dioxide and hydrogen, is a reversible exothermic reaction of particular interest due to its industrial importance. It is a step in many industrial processes, particularly ammonia and hydrogen production.

Two main classes of catalysts are used industrially as shift catalysts; iron-based catalysts and copper-based catalysts. The iron-based catalysts are some of the earliest heterogeneous catalysts used industrially and are commonly called high temperature shift catalysts operating in the temperature range of about 320-450°C. Most industrial high temperature shift catalysts contain chromium oxide as well as iron oxide and it is generally believed that Cr_2O_3 acts as a stabiliser, retarding sintering and loss of surface area of the iron oxide. The optimum beneficial Cr_2O_3 content appears to be about 14 wt% Cr_2O_3 [1], but typically commercial high temperature shift catalysts contain 8-10% Cr_2O_3. Before use the catalyst, which is essentially Fe_2O_3 as manufactured, has to be reduced to Fe_3O_4 which is thought to be the active phase.

Numerous studies of the kinetics and catalysts of the water-gas shift reaction over iron oxide/chromium oxide high temperature shift catalysts have been reported in the past 20 years, but there is little or no agreement as to either the precise form of the rate equation or the values of the rate constants, or surprisingly the activation energy. Some workers have proposed first order kinetics [2,3], others

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second order [4] to explain the conversions, but many others [5-9] have suggested more complex and different rate equations.

Bohlbro [10] believes that much of the confusion is due to impurities in the reacting gases, such as H_2S. Both Bohlbro and Ruthven [11] also attribute confusion over rate expressions to the kinetic data being taken under diffusional limitations. It has also been experimentally observed that catalysts from different manufacturers give different rate equations even under identical conditions [12], suggesting that the kinetics are dependent on the method of catalyst preparation.

Podolski and Kim [13] studied the kinetics at atmospheric pressure in a recycle reactor and attempted to discriminate statistically between most of the previously proposed kinetic expressions. They found that only the Langmuir-Hinshelwood and the power-law models could adequately describe the reaction behaviour. Fott et al. [14] noted that 20 different kinetic equations had been proposed in the last 40 years, differences among authors being especially in their opinions of the reaction mechanism and the effect of pressure on the reaction rate. They suggest that the reasons for this are that most kinetic measurements were made at atmospheric pressure only and using integral rather than differential reactors. Data from their differential reactor at pressures up to 10 bar also favour a Langmuir-Hinshelwood model with the surface reaction between adsorbed CO and H_2O the limiting step.

Measured activation energies range from 40.6 kJ mol^{-1} [4] through 100 kJ mol^{-1} [7,13], 115 kJ mol^{-1} [15], 134 kJ mol^{-1} [3], 142 kJ mol^{-1} [17] to 150 kJ mol^{-1} [18]. As Ruthven [11] points out, the low values are probably affected by diffusion limitations and the values obtained by workers using small catalyst particles are generally in the range 100-113 kJ mol^{-1}.

All workers [2,4,5,14,19] have reported that the rate of reaction increases with increase in pressure. Fott [14] asserts that the effect of pressure is the main criterion which enables the preliminary suitability of a given kinetic equation to be evaluated. His experimental data show an increase in rate with pressure up to a pressure of about 5 bar; further increase in pressure had little effect on the rate. Chandra [19] also reports that the rate doubles as the pressure increases from 5 to 15 bar, but then increases only slightly as the pressure was further increased. Padovani and Lotteri [20] found that the rate increased to a maximum at 10-20 bar and decreased with further increases in pressure.

The fall of catalyst activity with time has been studied by numerous workers. Mars [3] found that the catalyst loses activity due to sintering, the degree of which is a function of temperature and the age of the catalyst. For catalyst which had been used at temperatures of 510-550°C, the loss in activity agreed well with the findings of Hoogschagen and Zwietering [21]. They tried to establish a quantitative relationship between the decrease in surface area of a catalyst and its age at different operating temperatures as follows:

$$\frac{1}{A^n} - \frac{1}{A_o^n} = Kt \exp(-E/RT)$$

where A_o = specific surface area of a fresh catalyst, K is a constant, E is activation energy of sintering, t is time and n is an experimentally determined constant.

Chandra et al. [22] made experimental observations which were almost in agreement with Mars and Hoogschagen. They found a fast rate of fall in activity during the initial period up to 400 h, less fast between 400 and 700 h and after 1000 h the activity became more or less constant.

In summary, the best evidence seems to suggest that the reaction over commercial sized catalyst suffers from diffusional limitations, that the kinetics are best explained by a Langmuir-Hinshelwood model, with the surface reaction rate between adsorbed CO and H_2O being limiting, and that the catalyst ages due to sintering initially at a very fast rate but the activity becomes almost constant after a few months.

In this paper, the importance of mass transfer effects in the reaction over a commercial catalyst is assessed and other aspects of the reaction kinetics are described in the succeeding papers. Two types of integral reactor were used. A pilot plant with a large tubular reactor containing two sizes of standard catalyst pellets simulated the conditions in full-scale reactors. A pressure micro-reactor unit allowed the use of smaller sizes of catalyst particles and shorter reaction times, which were needed to study some features of catalyst performance, e.g. intrinsic activity and initial decay of activity.

EXPERIMENTAL

Semi technical unit

A simplified line diagram for the semi technical unit is shown in Figure 1. The reactor is mounted vertically with downward gas flow. The reactor dimensions are 2 inch ID by 60 inch length from flange face to flange face and it is made from 18/8/Ti stainless steel tube. Its maximum working pressure is 100 bar at 600°C. It is fitted with a thermocouple sheath of the same material, some 59 inches in length and 1/4 inch ID. The design allows the sheath to seat in the top cover where a seal is made with a Klingerite gasket. The threaded end of the sheath then projects through the top cover and is held in place with a nut. The reactor is heated by five pyrobar heaters of the following ratings (from the top of the reactor), ½, 2, 2, 1, ½ kW. The reactor length occupied by each heater is as follows; 6, 12, 12, 12 and 6 inch, leaving some 6 inches of reactor free at each end from the flange faces.

The catalyst and packing are supported on the metal stool some 7 inches long standing on the flange face of the bottom cover. A hole in the upper face of this

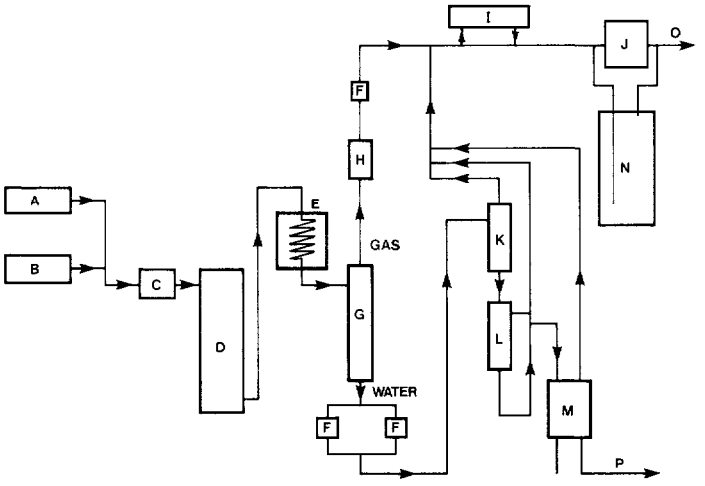


FIGURE 1 Simplified line diagram of semi-technical unit. A, mixed gas supply; B, de-ionised water supply; C, vapouriser; D, reactor; E, cooler; F, filter; G, high pressure separator; H, demister; I, chromatograph; J, gas meter; K, low pressure separator; L, decanter; M, storage vessel; N, lute; O, line to vent; P, line to drain.

circular stool allows the thermocouple sheath to pass through and locates it. The normal catalyst charge is 350 ml which occupies about 7 inches with the bed exit some 12 inches from the bottom flange face. The space above the bed is packed with fired alumina pellets up to the point where the line from the vaporiser enters. The vaporiser is located horizontally adjacent to the reactor and the line from the vaporiser is welded to the reactor some 9 inches from the top flange face. Thermocouples are positioned in the sheath to measure temperatures at the catalyst bed inlet, mid point and exit.

The gases exit the reactor leave through a high pressure line welded to the reactor some 6 inches from the bottom flange face and proceed to the cooler, an enclosed cylindrical shell through which cold water is continuously passed. The cooled wet gas then passes to the HP separator from which gases exit from the top and pass to a lute protected gas meter. The gas is analysed by process gas chromatograph. Water exits from the bottom of the HP separator, passes through an LP separator to a decanter/measuring vessel before being drained.

A standard gas from a battery was used for the experiments containing nominally 10% CO, 10% CO_2 and 80% H_2. The catalysts were reduced in a mixture of battery gas and steam (steam ratio 1.0) and then operated as close to isothermality as possible at temperatures in the range 310 to 450°C.

Microreactor unit

In the microreactor unit, horizontal stainless steel tubes of different diameters were used as reactors. The reactor heating system consisted of two brass

blocks, each with channels which, when the two blocks were fitted together, formed cylindrical holes through the block, of diameter 3.18, 4.75, 6.35 and 7.94 mm. The reactor tube was fitted into the appropriate cylindrical hole. Reactors of 4.75 mm internal diameter were used in this work. Both blocks were heated with a cartridge heater set into further holes at each end of the block. The temperatures of the blocks were controlled by thermocouples in 1.6 mm holes in the block. Temperatures at the catalyst bed inlet and exit were measured by the fixed thermocouples and another thermocouple, lying along the channel next to the tube, enabled temperature profiles to be carried out. The whole assembly, with the steam generator, was installed in an insulated box on the top of an oven containing the sample valves.

The gas feed for the unit was taken from cylinders containing a mixture of composition, 80% H_2/10% CO/10% CO_2. The gas rate was calculated from the pressure drop in a manometer across a length of capillary tubing. The manometer was calibrated by the use of a bubble flowmeter connected to unit outlet. Separate calibrations were needed for each process gas composition and pressure.

The steam feed to the reactor was obtained by the passage of a measured flow of de-ionised water, first through a steam generator and then a capillary into the gas feed. The 75 ml water reservoir was under nitrogen pressure and the water rate was measured directly from the fall in a graduated sight glass after appropriate isolation. The steam generator consisted of a length of 1/4 inch o.d. stainless steel tubing packed with stainless steel Dixon rings and fitted into an electrically-heated cylindrical brass block. A bulkhead Drallim coupling fitted with a sintered bronze disc was installed in the water supply line just before the steam generator. This device filtered out particles which would have otherwise blocked the capillary tube and it also gave a much smoother water flow, thus reducing any tendency for the steam supply to be irregular.

Gases were analysed by on-line gas chromatography, the feed gas directly and the exit gas after passage through an ice-cooled trap and a U-tube containing anhydrous calcium sulphate to dry the gas. The primary column consisted of 3.05 m of Porapak 'T' which split and held back CO_2 and water vapour from CO and H_2. Two secondary columns, 3.05 m of Porapak 'S' and 3.66 m of molecular sieve 13X, were used in parallel with appropriate flow switching. The columns were in an oven at 60°C. The CO and H_2 were separated on the molecular sieve column, whereas the CO_2 was passed down the Porapak 'S' column. Residual water vapour was backflushed from the primary column. A Taylor Servomex microkatharometer was used as detector.

Catalyst

The catalyst used in the semi technical tests was standard ICI catalyst 15, in both 15-4 (8.5 mm x 11.0 mm pellets) and 15-5 (5.4 mm x 3.6 mm pellets) forms. Similar standard 15-5 catalyst (ex drum 206) was crushed and sieved to 10-12 mesh size (1.68 - 2.00 mm) for most runs in the microreactor unit. Also some runs were

TABLE 1
Values of rate constant k/s^{-1}

Operating pressure /bar		16		21		31	
Size of catalyst (S = 5.4 x 3.6 mm; L = 8.5 x 11.0 mm)		S 15-5	L 15-4	S 15-5	L 15-4	S 15-5	L 15-4
Temperature /°C	310	0.019	0.018	0.018	0.018	0.017	0.017
	330	-	0.041	0.046	0.044	0.043	0.043
	350	0.103	0.090	0.114	0.091	0.092	0.097
	370	-	0.156	0.216	0.139	0.190	0.154
	400	0.461	0.334	-	-	0.404	0.256
	425	-	-	-	-	0.586	0.358
	450	1.200	0.829	-	-	0.918	0.565

carried out with whole pellets (which needed a slight reduction in diameter to fit the microreactor tube) and some runs used catalyst crushed and sieved to 22-25 mesh size (0.60-0.70 mm). In two series of runs on the microreactor the beds were diluted 6x with fused α-alumina particles of equivalent size.

RESULTS

Semi technical plant

In order to investigate the extent of diffusion effects in the semi-technical reactor, runs were carried out with 350 ml of standard catalyst in two different pellet sizes (5.4 x 3.6 mm and 8.5 x 11.0 mm) over the temperature range 310 to 450°C. Reasonable isothermality of the bed was possible at temperatures of up to 400°C. All of the data were obtained after the catalysts had been operated continuously for some 2 weeks. The dry gas feed rate was 1.75 m^3 gas (measured at 20°C and 1 bar) h^{-1} and operation was confined to a steam ratio (volume steam/volume dry gas) of 1.0. The total pressure of operation was varied in the range 16-31 bar. From the experimental data obtained it is possible to calculate psuedo first order rate constants assuming kinetics of the form:

$$\frac{dp_{CO}}{dt} = k\,(p_{CO} - p_{CO\ equilibrium})$$

or in integrated form:

$$k = S_v \text{ in } (1/(1 - C/C_e))$$

TABLE 2

CO conversions in microreactor unit. Catalyst bed undiluted and of 10-12 mesh size unless otherwise stated.

Catalyst bed length /cm	Catalyst volume /ml	% CO conversion at various temperatures /°C 350	310	330	350
11.2	2.00	50.9	-	80.3[d]	49.4
2.54	0.453	34.3	9.2	18.5	33.0
5.08	0.905	43.1	18.6	30.2	46.7
10.2	1.81	55.8	18.1	33.0	51.0
15.2	2.72	54.8	11.8	30.8	50.5
2.54	0.45[a]	49.8	15.4	30.3	46.9
15.2	0.45[a,c]	54.9	19.0	37.2	-
2.54	0.45[b]	51.6	16.9	28.4	45.5
15.2	0.45[b,c]	54.2	13.5	27.6	50.5

[a]whole pellets of catalyst
[b]catalyst particles 22-25 mesh size
[c]catalyst bed diluted
[d]run at 400°C

where k = rate constant (s^{-1}), S_v = space velocity at reaction conditions (s^{-1}) both defined in terms of empty bed volume, C = % conversion of CO, and C_e = % conversion of CO at equilibrium.

Table 1 shows the values of k obtained for the two different sized catalysts over the range of temperatures and pressures examined.

Microreactor unit

In runs with the first catalyst charge, a high initial activity, which decayed rapidly, was observed. This change in activity, which occurs too rapidly to be observed on larger units, is described in the following paper [26]. Details of several series of runs with different catalyst charges are given in Table 2. All runs were at 31.6 bar pressure, a gas space velocity (STP) of 5000 h^{-1} and a dry gas/steam ratio of unity. Tests with a given catalyst charge were carried out in the temperature sequence in Table 2. The first row of results in Table 2 was obtained with catalyst which had reached a steady state activity after a long series of runs under varied conditions. The two values of CO conversion at 350°C

are the same and not significantly different from the values obtained in the semi technical unit under comparable conditions. The stabilising procedure described elsewhere [26] was used before all the other series of runs. Comparison of the first and second runs at 350°C in these series shows that the stabilising procedure almost, but not quite, produced the steady state activity.

In the second series of runs, the catalyst bed length was varied under otherwise constant conditions. The effects of change in catalyst particle size and of bed dilution were examined in the third series of runs. Two main conclusions can be drawn from the results in Table 2. First, the CO conversions per unit volume of catalyst in runs with short catalyst beds (2.54 cm and possibly also the 5.08 cm bed) are lower than those obtained with catalyst beds ≥ 10 cm long. The origin of this fall is discussed below and gives a limit to the operation of the micro-reactor unit. Second, a comparison of the runs with catalyst beds of length 10.2 or 15.2 cm shows that neither variation in catalyst particle size (from 0.6 to 5 mm) nor catalyst bed dilution has any significant effect on CO conversions.

MASS TRANSFER EFFECTS ON REACTION

Semi technical plant

Film Diffusion. It is necessary to consider first the possibility of a film diffusion limitation in the semi technical plant. A common method of checking for this is to keep the space velocity constant and vary the flowrate noting whether the conversion remains constant. Constant conversion under these conditions is taken to indicate the absence of external mass transfer influence. This test has been criticised by Chambers and Boudart [28] since at the low particle Reynolds numbers encountered in test units, heat and mass transfer coefficients are relatively insensitive to changes in flow rates.

The possibility of gas film limitation can be estimated by assessing the magnitude of the concentration gradient of the reactants between the bulk fluid and the surface of the catalyst particle. The most simplified example of this procedure is due to Wheeler [29] who has calculated that for a completely diffusion limited reaction:

$$k_\infty = 10\ [V_L/(Ma^3\ P_T)]^{\frac{1}{2}}$$

where k_∞ = first order rate constant/unit volume reactor (s^{-1}), V_L = linear velocity of gas through empty reactor (cm s^{-1}), M = average molecular weight of gas, a = pellet size (cm), P_T = total pressure (bar). If the experimentally measured first order rate constant is less than 10% of the calculated k_∞ then film diffusion limitation is unlikely. Table 3 compares values of the experimental first order rate constant with calculated values for k_∞ at the highest operating pressure used (31 bar) and it can be seen that a significant film diffusion limitation is

TABLE 3
Comparison of experimental $k(s^{-1})$ with calculated k_∞ (s^{-1}) at 31 bar pressure, 1.75 nm^3 h^{-1} dry gas, 1.0 steam ratio.

Catalyst size	15-5 5.4 x 3.6 mm	15-4 8.5 x 11.0 mm
Temperature/°C		
310 k_∞	5.6	2.0
k	0.02	0.02
350 k_∞	5.7	2.1
k	0.09	0.10
370 k_∞	5.9	2.1
k	0.19	0.15
400 k_∞	6.0	2.1
k	0.40	0.26
425 k_∞	6.1	2.2
k	0.59	0.36

unlikely.

Madon and Boudart [30] suggest that the most appropriate experimental criterion for the absence of transport influence on catalytic rates is the Koros-Nowak criterion. Briefly this states that a system is free from any transport influences if the rate of reaction per unit concentration of active material is constant; e.g. if the reaction rate per unit surface area of active material is constant. Measurements of total surface area and of the activity in the semi technical plant of both new and a plant discharged catalyst 15-4, suggest that the reaction rate/unit surface area is constant up to a temperature of about 350°C but not at higher temperatures. This is additional evidence that there is no film diffusion limitation in the semi technical plant at temperatures up to 350°C and in conjunction with the satisfaction of Wheeler's criterion at higher temperatures suggests that catalyst 15-4 experiences a pore diffusion limitation at temperatures above 350°C at 31 bar total pressure.

Effectiveness factors and instrinsic rate constants. Inspection of the experimental results reveals that the two differently sized catalysts give similar performance at temperatures up to about 360°C, but above this temperature the smaller pellets of 15-5 give higher conversions and calculated rate constants than the larger pellets of 15-4.

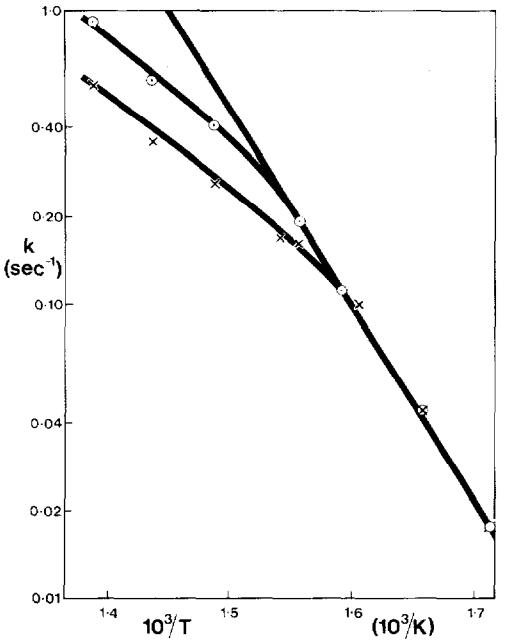


FIGURE 2 Arrhenius plot for ICI catalysts 15-4 and 15-5 at 31 bar and 1.0 steam ratio. X, catalyst 15-4; ⊙ , catalyst 15-5. Activation energies: 124.8 kJ mol^{-1} at low temperatures; 62.4 kJ mol^{-1} for catalyst 15-4 at high temperatures; 65.7 kJ mol^{-1} for catalyst 15.5 at high temperatures.

A plot of ln k against reciprocal temperature (Figure 2) for 15-4 and 15-5 at 31 bar operating pressure shows a bend at around 350-370°C and suggests that the reaction becomes significantly pore diffusion limited above 350°C for the large pellets and above 370°C for the smaller pellets. The slopes of the lines show a fall in apparent activation energy from approximately 125 kJ mol^{-1} to approximately 63 kJ mol^{-1} at the higher temperatures.

For a pseudo first order reaction, following the treatment of Wheeler [31]

$$k \text{ (observed rate constant/unit weight of catalyst) } cm^3\ g^{-1}\ s^{-1} = (S_x/V_p)^2 \frac{Deff}{\rho_p} h \tanh h$$

where S_x = geometric surface area of pellets (cm^2), V_p = volume of pellet (cm^3), Deff = effective diffusion coefficient ($cm^2\ s^{-1}$), ρ_p = pellet density (g cm^{-3}), h = modulus = $(V_p/S_x)(k_i\rho_p/Deff)^{\frac{1}{2}}$, k_i = intrinsic rate constant ($cm^3\ g^{-1}\ s^{-1}$). If it is assumed that k_i and Deff are the same for the two different sized catalysts then the ratio of the values of the modulus h for the two catalysts can be calculated.

$$h_{15\text{-}4}/h_{15\text{-}5} = 0.52$$

From the measured rate constants at any given temperature, since $k/k_i = E$

TABLE 4

31 bar operation

Temperature /°C	Catalyst	Observed rate constant /$cm^3\ g^{-1}\ s^{-1}$	h^a	E^a	Deff /$cm^2\ s^{-1}$	h^b	E^b	Relative activities of 15-4/15-5	Intrinsic rate constant /$cm^3\ g^{-1}\ s^{-1}$
310	15-5	0.013				0.15	0.99	0.98	0.013
	15-4	0.013				0.29	0.97		
330	15-5	0.033				0.25	0.98	0.96	0.035
	15-4	0.034				0.48	0.94		
350	15-5	0.071				0.35	0.96	0.93	0.080
	15-4	0.076				0.67	0.89		
370	15-5	0.146	0.65	0.90	0.0050			0.72	0.166
	15-4	0.108	1.25	0.65	0.0049				
400	15-5	0.311	1.10	0.70	0.0044			0.63	0.444
	15-4	0.202	2.15	0.44	0.0046				0.459
425	15-5	0.469	1.7	0.53	0.0039			0.59	0.885
	15-4	0.282	3.2	0.31	0.0042				0.910
450	15-5	0.827	2.1	0.45	0.0050			0.53	1.86
	15-4	0.445	4.1	0.24	0.0052				1.85

[a] calcualted from relative rate constants

[b] assuming Deff = 0.0050 from 310-370°C for 15-5

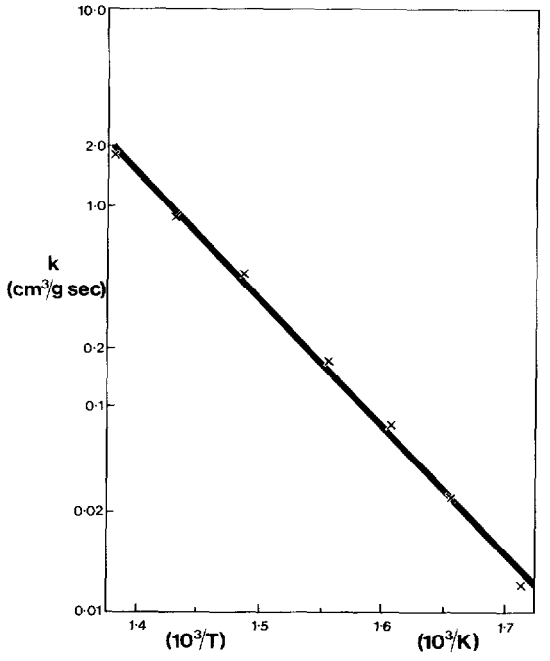


FIGURE 3 Arrhenius plot of the calculated intrinsic first-order rate constants for ICI catalyst 15. Activation energy, 121.8 kJ mol^{-1}.

(effectiveness factor) the ratio of the effectiveness factors for the two pellet sizes can also be calculated. Since E = tanhh/h, a trial and error method enables the actual values of E and h to be obtained.

Table 4 shows the rate constant data used and the values of E and h obtained in this way for 31 bar pressure operation. The rate constants used are those already quoted in Table 3 but corrected to a weight of catalyst basis by dividing by the relevant catalyst bulk densities. The above procedure for obtaining E and h is reasonably accurate down to temperatures of 370°C. From these calculated values of h and E, both k_i and Deff can be calculated and the values obtained are also shown in Table 4. If it is assumed that Deff does not vary with temperature in the range 310 to 370°C then values of h and E for temperatures in this range can be calculated and are also shown in Table 4. It is clear that the smaller pellets only show a significant advantage in activity at temperatures of 370°C and above.

Diffusion coefficient. The data suggest that the effective diffusion coefficient for these catalysts at 31 bar and 300-400°C is about 0.005 $cm^2\ s^{-1}$. Calculation of the bulk gas diffusion coefficient for CO in reaction gas at 31 bar pressure gives values of 0.035 $cm^2\ s^{-1}$ at 300°C and 0.052 $cm^2\ s^{-1}$ at 450°C using the methods described by Reid and Sherwood [32]. Calculation of the Knudsen diffusion coefficient gives values of 0.00044 r at 300°C and 0.00049 r at 450°C, where r is the pore radius in Angstom units. Typical mean pore radii for these catalysts are 50 Å freshly reduced rising to 300 Å in use giving Knudsen coefficients

TABLE 5

Effect of pressure

Temperature /°C	Catalyst	Pressure 21 bar			Pressure 16 bar		
		E	k /$cm^3 g^{-1} s^{-1}$	observed k /$cm^3 g^{-1} s^{-1}$	E	k /$cm^3 g^{-1} s^{-1}$	observed k /$cm^3 g^{-1} s^{-1}$
310	15-5	0.99	0.013	0.014	0.99	0.013	0.015
	15-4	0.97	0.013	0.015	0.98	0.013	0.015
330	15-5	0.98	0.034	0.035	0.98	0.034	
	15-4	0.95	0.033	0.037	0.96	0.034	0.034
350	15-5	0.97	0.078	0.088	0.98	0.078	0.079
	15-4	0.93	0.074	0.076	0.94	0.075	0.075
370	15-5	0.93	0.151	0.166	0.94	0.152	
	15-4	0.75	0.125	0.116	0.80	0.133	0.130
400	15-5	0.80	0.355		0.85	0.377	0.355
	15-4	0.51	0.234		0.56	0.257	0.278
425	15-5	0.60	0.531		0.66	0.584	
	15-4	0.38	0.346		0.42	0.382	
450	15-5	0.52	0.967		0.57	1.060	0.923
	15-4	0.29	0.537		0.34	0.629	0.691

of 0.022-0.132 $cm^2\ s^{-1}$ at 300°C and 0.025-0.147 $cm^2\ s^{-1}$ at 450°C. Since Deff = $(\theta/\tau)D$; and the porosity θ of these catalysts is close to 0.5, the following values of τ the tortuosity factor are calculated.

For bulk diffusion τ = 3.5 - 5.2 and for Knudsen Diffusion τ = 2.2 - 2.5 (for new catalyst; on used catalyst bulk diffusion would clearly predominate).

Values of τ of 1.85 - 2.05 have been reported for Fe_2O_3/Cr_2O_3 catalysts [33], whilst a value for Deff/D bulk of 0.054 was found for a water gas shift catalyst [34], which would give a value of τ of 9.3 if the porosity was 0.5.

Effect of pressure and temperature. Figure 3 shows an Arrhenius plot of the calculated intrinsic first order rate constants of the catalysts from which a true activation energy of 121.8 kJ mol^{-1} has been deduced. The effect of operating at lower pressures than 31 bar can be estimated since, if bulk diffusion is concerned, Deff is proportional to the reciprocal pressure. If k_i is assumed independent of pressure then h is proportional to $p^{\frac{1}{2}}$. Values of h for different pressures can then be calculated and the appropriate values of E (effectiveness factor) deduced. Table 5 shows calculated values of E and thence predicted values for the apparent rate constant under these conditions together with the observed rate constants where available (from Table 1). There is fair agreement between the observed and calculated values down to 16 bar pressure.

In summary, from experiments in an integral semi-technical reactor at pressures from 16-31 bar and temperatures from 310-450°C using a single inlet gas composition with an excess of steam so that the reaction can be taken to be pseudo-first order in carbon monoxide partial pressure the following conclusions can be drawn:
The true activation energy for the reaction over ICI catalyst 15 is 121.8 kJ mol^{-1}.

The reaction over 5.4 x 3.6 mm pellets (15-5) is not pore diffusion limited at temperatures of 370°C and below at pressures up to 31 bar. Over the larger 8.5 x 10.5 mm pellets (15-4) the reaction becomes pore diffusion limited at temperatures above 350°C at 31 bar pressure. A method of optimising an HT shift converter in the light of this information was the subject of a patent application [16].

The effective diffusion coefficient of CO in shift gas for these catalysts is approximately 0.005 cm^2 s^{-1} in the temperature range 370-450°C.

Microreactor

The results obtained with the microreactor unit are in good agreement with this analysis of catalyst performance in the semi-technical unit. No pore diffusion effects should be apparent with catalyst particles of size 0.6 to 5 mm in the temperature range 310-350°C and indeed no variation of CO conversion with particle size was found. The kinetic parameters provide further evidence. The CO conversions measured at 350°C over steady-state catalyst in the microreactor unit were not significantly different from the experimental values in the semi-technical unit under the same conditions. In a following paper [27] the activation energy for the reaction over steady-state catalyst in the microreactor is found to be 119.3 kJ mol^{-1}. Again this is in good agreement with the value of 121.8 kJ mol^{-1} calculated above for the rate of reaction free of mass transfer limitation.

Limits to microreactor performance. For the analysis of the microreactor results given above to be valid it is essential the observed performance should have been controlled solely by the properties of the catalyst and not influenced by inter-particle heat and mass transfer in the catalyst bed. An estimate of Reynolds number for gas flow between catalyst particles in the reactor under typical conditions gives a value of about 5. Under all the experimental conditions reported here the gas flow was well into the region of streamline flow, so standard expressions for diffusion and heat conduction can be used.

If radial heat transfer in the reactor is insufficient for the catalyst bed to reach an isothermal state then the higher temperature near the axis of the bed gives higher reaction rates than expected. With the assumption of cylindrical symmetry, the bed temperature at radius r can be given as:

$$T = T_0(1 + \alpha)$$

where T_0 = reactor wall temperature.

The specific rate constant applicable at r is then

$$k = A \exp(-E/RT) = A \exp(-E/T_0(1 + \alpha)) \qquad (1)$$

for small deviations from the isothermal state, $\alpha << 1$ and equation (1) can be approximated to:

$$k = A \left[1 + \frac{E}{RT_0} \alpha \right] \exp (E/RT_0)$$

The rate of reaction can be obtained by integration of a cylindrical element of thickness δr and length ℓ, where ℓ is short enough for conditions to be essentially constant along the reactor. For a first order reaction,

$$m = \pi r_0^2 \ell \, c \, A \exp \left[- \frac{E}{RT_0} \right] \left[1 + \frac{2}{r_0^2} \int_0^{r_0} \alpha \, r \, dr \right]$$

where m = rate of reaction r_0 = reactor radius c = concentration of reagent.

The rate of reaction under isothermal conditions, m_0, is given by

$$m_0 = \pi r_0^2 \ell \, c \, A \exp \left[- \frac{E}{RT_0} \right]$$

The fractional increase in rate, $\emptyset$, due to a non-isothermal state is then

$$\emptyset = \frac{m}{m_0} - 1$$

$$\emptyset = \frac{2E}{r_0^2 RT} \int_0^{r_0} \alpha r dr$$

Now the standard equation for heat conduction in a cylinder [23] can be applied.

$$\alpha T_0 = \frac{H_0}{4\lambda} (r_0^2 - r^2)$$

where H_0 = rate of heat generation per unit volume and λ = thermal conductivity of catalyst bed.

Hence:

$$\emptyset = \frac{EH_0}{2r_0^2 R\lambda T_0^2} \int_0^{r_0} (r_0^2 r - r^3) dr$$

$$\phi = \frac{EH_0 r_0^2}{8R\lambda T_0^2} \qquad (2)$$

The increase in temperature at the axis of the reactor is:

$$T_0\alpha_0 = \frac{H_0 r_0^2}{4\lambda} \qquad (3)$$

Equations (2) and (3) can be applied to typical reaction conditions described above. The effects of limited heat transfer are more pronounced at the highest temperatures because the rate of reaction, and hence the rate of heat evolution, is maximal. The parameters for reaction at 350°C (except λ) can be derived from the results above and the standard enthalpy change at that temperature, 38.58 kJ mol^{-1}. The principal source of error is the estimate of λ, which is not known for a bed of high-temperature shift catalyst. Experimental values for catalyst and other granular packed beds [24,25] suggest that λ lies in the range 21 x 10^{-4} to 12.5 x 10^{-3} J cm^{-1} s^{-1} $°C^{-1}$.

The values of the temperature rise in the centre of the bed in the inlet region and the consequent increase in reaction rate are then in the ranges 0.7 to 4°C and 1% to 8% respectively. These values are small enough for the approximations made in the calculations above to be valid. It is likely therefore that the microreactor was operated in a region where errors due to heat transfer limitations were not significant, although this state was approached under the most severe conditions. An experimental test for these errors lies in the comparison of conversions over diluted and undiluted catalyst beds, giving different values of H_0, at the same space velocity and temperatures (Table 2). The results show no decrease in conversion on dilution of the catalyst bed thus confirming the estimated limits.

In contrast the calculation indicates that use of the microreactor with this reaction above 350°C or with larger diameter reactor tubes would lead to significant errors as indeed would also occur with a faster or more exothermic reaction under the same experimental conditions.

Radial mass transfer takes place by diffusive mixing only in streamline flow. A simple measure of the efficiency of radial mixing between catalyst particles is given by a comparison of nominal residence time with the diffusion time corresponding to the radius of catalyst particles given by the Einstein equation:

$$t_c = r_c^2/2D$$

where D is the diffusion coefficient for CO in the gas mixture, estimated above

to be 0.035 $cm^2\ s^{-1}$ at 350°C. The nominal residence time in the catalyst bed was typically 10 s whereas the diffusion time is calculated to be about 0.15 s. Thus, radial mixing is sufficently fast, which is to be expected as the reaction regime in the catalyst itself is essentially free of pore diffusion limitations.

The fast diffusional mixing indicates that axial mass transfer, not usually a source of error in experimental catalytic reactors, could be of significance in short microreactor beds. The following treatment gives an analytical solution for a first order reaction. The rate equation is:

$$-\frac{dc}{dt} = kc$$

and a mass balance across an elemental disc of thickness $\delta\ell$ of the catalyst bed gives:

$$D\pi r_o^2 \frac{d^2c}{dt^2} - v\frac{dc}{d\ell} - \pi r_0^2 kc = 0 \qquad (4)$$

Transferred to the dimensionless co-ordinates, $y = c/c_0$ and $x = \ell/L$ and real space velocity S equation (4) becomes:

$$\frac{d^2y}{dx^2} - 2C_1\frac{dy}{dx} - 2C_2y = 0 \qquad (5)$$

where $C_1 = SL^2/2D$ and $C_2 = kL^2/2D$.

With use of appropriate boundary conditions, equation (5) can be solved [35] to give equation (6).

$$y_1 = \frac{4C_3\exp\,[C_1(1-C_3)]}{[(1+C_3)^2 - (1-C_3^2\exp\,(-2C_1C_3)]} \qquad (6)$$

where $C_3 = (1 + 2C_2/C_1^2)^{\frac{1}{2}}$

The fractional conversion is:

$$\theta = 1 - y_1 \qquad (7)$$

Equation (6) gives limits

$y_1 = \exp\,(-k/S)$ as $L^2/D \to \infty$

corresponding to an integral reactor, and

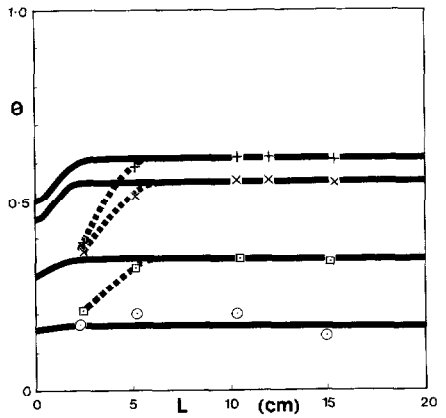


FIGURE 4 Experimental and theoretical variation in conversion (θ) with microreactor bed length (L). ⊙ , 310°C; ⊡ , 330°C; +, initial 350°C; x, final 350°C; —— , theoretical for ideal mixing.

$$y_1 = 1/(1 + k/S) \text{ as } L^2/D \to 0$$

corresponding to a continuous stirred tank reactor.

An application of equations (4) and (7) to the experimental results on the water-gas shift reaction requires the use of conversion of CO towards the equilibrium value. The values of k are taken from the limiting values of maximum L; values of D were estimated for the gas phase above. The theoretical fall in θ at small L is compared with the experimental results in Figure 4. The experimental values of θ are independent of bed length for L ≥ 10 cm and so the other results in this and the following papers were obtained with bed lengths of 10-15 cm. The conversions, except for the results at the lowest temperature, fall with decreasing catalyst bed length rather more rapidly than calculated from the model alone. Thus, for the 350°C results the model predicts a significant fall at about 1.5 cm bed length whereas the comparable experimental fall occurs at about 5 cm bed length. In spite of this discrepancy, the theoretical model of axial mass transfer provides a satisfactory basis for microreactor design when the reactor bed length is kept at least several times the calculated minimum.

The cause of the differences between theoretical conversions at short bed lengths was probably some degree of by-passing in addition to axial diffusive mixing. Refinement of the model to allow for the presence of the solid catalyst particles seems unlikely to make the necessary changes. Values of D would need to be larger by about an order of magnitude to get optimum fit and it is improbable that the estimate of D is so far in error. Moreover some of the experimental values of θ fall below the minimum limiting values (i.e. for CSTR) calculated for axial mass transfer and this is difficult to interpret other than by an assumption of some by-passing.

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NOMENCLATURE

T = temperature in reactor bed (°C)
T_0 = reactor wall temperature (°C)
θ = deviation from isothermal state (-)
A = pre-exponential factor of first-order rate constant (s^{-1})
E = activation energy for first-order reaction (kJ mol^{-1})
r_0 = reactor bed radius (cm)
ℓ = length of cut element of catalyst bed (cm/axial co-ord of cb)
c = concentration of reagent
r = radial co-ordinate of cb (cm)
m = rate of reaction (mol s^{-1})
m_0 = rate of reaction under isothermal conditions (mol s^{-1})
$\emptyset$ = fractional increase in rate due to non-isothermal conditions (-)
H_0 = rate of heat generation by reaction (J cm^3 s^{-1})
x = thermal conductivity of catalyst bed (J cm^{-1} s^{-1} $°C^{-1}$)
t = time (s)
t_c = diffusion time between catalyst particles (s)
r_c = radius of catalyst particles (cm)
D = diffusion coefficient for CO in gas mixture (cm^2 s^{-1})
V = gas flow through catalyst bed (cm^3 s^{-1})
y = dimensionless reagent concentration (-)
x = dimensionless reactor bed length (-)
C_1 = $SL^2/2D$
C_2 = $kL^2/2D$
S = real space velocity (s^{-1})
n_1 = $C_1 (1 + C_3)$
n_2 = $C_1 (1 - C_3)$
C_3 = $(1 + 2C_2/C_1^2)^{\frac{1}{2}}$
C_4 = integration constant
C_5 = integration constant
y_1 = exit reagent concentration (-)
α = dimensionless conversion (-)

REFERENCES

1 M.I. Markina, G.K. Boreskov, F.P. Ivanovskii and B.G. Lyudkovskaya, Kinet. Katal., 2 (1961) 867.
2 K. Atwood, M.R. Arnold and E.G. Appel, Ind. Eng. Chem., 42 (1950) 1600-1602.
3 P. Mars, Chem. Eng. Sci., 14 (1961) 375.
4 J.M. Moe, Chem. Eng. Prog., 58(3), (1962) 33.

5 H. Bohlbro, J. Catal., 3 (1964) 207.
6 N.V. Kulkova and M.I. Temkin, Zh. Fiz. Khim., 23 (1949) 695.
7 V.I. Atrosschenko and B. Bibr, J. Appl. Chem., USSR, 1020 (1959).
8 H.M. Hulburt and G.D. Srini Vasan, AIChE J., 7 (1961) 143.
9 V.K. Puri, H. Mahapatra, R.M. Kursetji, N.C. Ganguly and S.P. Sen, Technol., 10 (1973) 224.
10 H. Bohlbro: An investigation on the kinetics of the conversion of carbon monoxide with water vapour over iron oxide based catalysts, 2nd ed. Gjellerup, Copenhagen, 1969.
11 D.M. Ruthven, Can. J. Chem. Eng., 47 (1969) 327.
12 F. Goodridge and H.A. Quazi, Trans. Inst. Chem. Eng., 45 (1967) T.274.
13 W.F. Podolski and Y.G. Kim, Ind. Eng. Chem. Process Des. Dev., 13 (1974) 415.
14 P. Fott, J. Vosolsobe and V. Glaser, Collection Czechoslov. Chem. Commun., 44 (1979) 652.
15 H. Bohlbro, Acta. Chem. Scand., 15 (1961) 502.
16 G.C. Chinchen, BP 1,500,089, USP 4,142,988.
17 G.G. Shchibrya, N.M. Morozov and M.I. Temkin, Kinet. Katal., 6 (1965) 1057.
18 F.P. Ivanovskii, G.E. Bruade and T.A. Semenova, Kinet. Katal., 5 (1964) 563.
19 M. Chandra, S.S. Singh, B. Banerjee, A.K. Sinha and N.K. Sinha, Technology, 9 (1972) 358.
20 C. Padovani and A. Lotteri, J. Soc. Chem. Ind., 56 (1937) 311 T.
21 J. Hoogschagen and P. Zwietering, J. Chem. Phys., 21 (1953) 2224.
22 M. Chandra, S.S. Singh, A.K. Sinha, B. Banerjee and S.R. Ghosal, Technology, 10 (1973) 208.
23 H.S. Carslaw and J.C. Jaeger, Conduction of Heat in Solids, 2nd edition, Clarendon Press, Oxford, 1959, p.191.
24 C.N. Satterfield, Mass Transfer in Heterogeneous Catalysis, M.I.T. Press, Massachusetts, 1970, p.170.
25 R. Krupiczka, Intern. Chem. Eng., 7 (1967) 122.
26 G.C. Chinchen, R.H. Logan and M.S. Spencer, Following paper (Part II).
27 G.C. Chinchen, R.H. Logan and M.S. Spencer, Following paper (Part III).
28 R. Chambers and M. Boudart, J. Catal., 6 (1966) 141.
29 A. Wheeler, Catalysis, vol. 2, chapter 2.
30 R. Madon and M. Boudart, Ind. Eng. Chem. Fundam., 21 (1982) 438-447.
31 A. Wheeler, Adv. Cat. Vol. 3.
32 R.C. Reid and T.K. Sherwood, The properties of gases and liquids, McGraw-Hill (1958).
33 E.L. Furen, D.V. Gernet, T.A. Semenova and M.P. Shmigirovskaya, Kinetika i Kataliz, 12, 5, (1971) 1283-9.
34 C. Bokhoven and H. Hoogschagen, J. Chem. Phys., 21 (1953) 159.
35 J.J. Carberry, "Chemical and Catalytic Reaction Engineering", McGraw-Hill, New York, 1976, p.148.